Meal modulation of circulating interleukin 18 and adiponectin concentrations in healthy subjects and in patients with type 2 diabetes mellitus.

BACKGROUND: A single high-fat meal induces endothelial activation, which is associated with increased serum concentrations of inflammatory cytokines. OBJECTIVE: We compared the effect of 3 different meals on circulating concentrations of interleukin 8 (IL-8), interleukin 18 (IL-18), and adiponectin in healthy subjects and in patients with type 2 diabetes mellitus. DESIGN: Thirty patients with newly diagnosed type 2 diabetes and 30 matched, nondiabetic subjects received the following 3 isoenergetic (780 kcal) meals separated by 1-wk intervals: a high-fat meal; a high-carbohydrate, low-fiber (4.5 g) meal; and a high-carbohydrate, high-fiber meal in which refined-wheat flour was replaced with whole-wheat flour (16.8 g). We analyzed serum glucose and lipid variables and serum IL-8, IL-18, and adiponectin concentrations at baseline and at 2 and 4 h after ingestion of the meals. RESULTS: Compared with nondiabetic subjects, diabetic patients had higher fasting IL-8 ($P < 0.05$) and IL-18 ($P < 0.01$) concentrations and lower adiponectin concentrations ($P < 0.01$) at baseline. In both nondiabetic and diabetic subjects, IL-18 concentrations increased and adiponectin concentrations decreased ($P < 0.05$) from baseline concentrations after consumption of the high-fat meal. After consumption of the high-carbohydrate, high-fiber meal, serum IL-18 concentrations decreased from baseline concentrations ($P < 0.05$) in both nondiabetic and diabetic subjects; adiponectin concentrations decreased after the high-carbohydrate, low-fiber meal in diabetic patients. IL-8 concentrations did not change significantly after consumption of any of the 3 meals. CONCLUSIONS: This study provides evidence that circulating IL-18 and adiponectin concentrations are modulated by familiar foodstuffs in humans. Meal modulation of cytokines involved in atherogenesis may represent a safe strategy for ameliorating atherogenetic inflammatory activity in diabetic patients.